Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

ÐµÐ¶Ðµ ÑĤÐ°Ðº, Ð´Ð°Ð²Ð°Ð¹ÑĤÐµ ÑĢÐµÑĪÐ¸Ð¼ Ð·Ð°Ð´Ð°Ñї Ñĥ Ð¿Ð¾ÑĪÐ°Ð³Ð¾Ð²Ð¾.

ÐłÐ±Ð¾Ð·Ð½Ð°Ñї Ð¸, ÑģÐºÐ¾Ð»ÑĮÐºÐ¾ Ð²Ð°Ðº Ñĥ ÑĥÐ¼ÐµÐ½Ñ-Ð¿ÑĢÐ¸Ð½Ð°Ð´Ð»ÐµÐ¶
Ð½Ð¾ÑģÑĤÐµÐ¹ Ñĥ ÐľÐµÐ»Ð°Ð½Ð¸ Ð² Ð½Ð°Ñ‡Ð°Ð»Ðµ âĢĶ x **.

1. **ÐĹÐµÐ»ÐµÐ½ÑĭÐ¹ Ð´Ð¾Ð¼:** Ð¾Ð½Ð° Ð¿ÑĢÐ¾Ð´Ð°Ð»Ð° ÑĤÑĢÐµÑĤÑĮ Ð² Ð·ÐµÐ»ÐµÐ½Ð¾Ð¼ Ð´Ð¾Ð¼Ðµ âĢĶ x/3
âĨĴ Ð¾ÑģÑĤÐ°Ð»Ð¾ÑģÑĮ: x - x/3 = (2x)/3

2. **ÐļÑĢÐ°ÑģÐ½Ñ‹Ð¹ Ð´Ð¾Ð¼:** Ð¿ÑĢÐ¾Ð´Ð°Ð»Ð° 2 Ð±Ð¾Ð»ÑĮÑĪÐµ, ÑĩÐµÐ¼ Ð² Ð·ÐµÐ»ÐµÐ½Ð¾Ð¼ Ð´Ð¾Ð¼Ðµ. Ð¢Ð°ÐºÐ¸Ð¼ Ð¾Ð±Ñĸ Ð°Ð¼ Ð¾ÑģÑĤÐ°Ð»Ð¾ÑģÑĮ (2x)/3 - (x/3 + 2) = (2x)/3 - x/3 - 2 = (x/3) - 2

3. **ÐŀÑĢÐ°Ð¶ÐµÐ²Ñ‹Ð¹ Ð´Ð¾Ð¼:** **Ð¿ÑĢÐ¾Ð´Ð°Ð»Ð° Ð¾ÑģÑĤÐ°Ð²ÑˆÑĥÑİÑģÑı Ñĩ Ð¾ÑģÑĤÐ°Ð²ÑˆÐµÐ¹ÑģÑı Ñĩ Ð¾ÑģÑĤÐ°Ð²ÑˆÐµÐ¹ÑģÑı** Ð² Ð°Ð¿ÐµÐ»ÑĮÑģÐ¸Ð½Ð¾Ð²Ð¾Ð¼ Ð´Ð¾Ð¼Ðµ.

ÐĴÐ¾ Ð·ÐµÐ»ÐµÐ½Ð¾Ð¼ Ð´Ð¾Ð¼Ðµ Ð¾Ð½Ð° Ð¿ÑĢÐ¾Ð´Ð°Ð»Ð° Ð¾ÑģÑĤÐ°Ð²ÑˆÑĥÑİÑģÑı Ñĩ Ð¾ÑģÑĤÐ°Ð²ÑˆÐµÐ¹ÑģÑı x/3, Ð·Ð½Ð°ÑĩÐ¸Ñĩ, Ð²
ÐºÐ¾Ð½Ñģ Ð½Ð¾Ð¼Ð¾ Ð´Ð¾Ð¼Ðµ Ð¿ÑĢÐ¾Ð´Ð°Ð»Ð° x/3 + 2

âĨĴ Ð¾ÑģÑĤÐ°Ð»Ð¾ÑģÑĮ:
(2x)/3 - (x/3 + 2) = (2x)/3 - x/3 - 2 = (x/3) - 2

3. **ÐłÐ°ÑĢÐ¶ÐµÐ²Ñ‹Ð¹ Ð´Ð¾Ð¼:** **Ð¿Ð¾Ð»Ð¾Ð²Ð¸Ð½Ñĭ ÑĤÐ°Ð¼ Ð½Ðµ Ð¾ÑģÑĤÐ°Ð»Ð¾ÑģÑĮ**.

âĨĴ Ð¿ÑĢÐ¾Ð´Ð°Ð»Ð°: (1/2) ÃĹ [(x/3) - 2]
âĨĴ Ð¾ÑģÑĤÐ°Ð»Ð¾ÑģÑĮ: (1/2) ÃĹ [(x/3) - 2]

ÐŁÐ¾ ÑĤÑģÐ»Ð¾Ð²Ð¸Ñİ, Ð¾ÑģÑĤÐ°Ð²ÑˆÐ¸Ðµ **5** Ð²Ð°Ðº Ñĥ ÑĤÐ¾Ñĩ Ð²Ð¾ÑģÑĤÐ¾Ðº.

Ð¢Ð°Ðº ÑĩÑĤÐ¾:
$$
\frac{1}{2}\left(\frac{x}{3} - 2\right) = 5
$$

ÐłÐµÑĪÐ¸Ð¼ ÑĥÑĢÐ°Ð²Ð½ÐµÐ½Ð¸Ðµ:

$$
\frac{x}{3} - 2 = 10
$$

ÐŁÑĢÐ¸Ð±Ð°Ð²Ð¸Ð¼ 2:
$$
\frac{x}{3} = 12
$$

ÐĴÑĭÑĩÐ¸ÑģÐ»Ð¸Ð¼ x:
$$
x = 36
$$

ÐłÐµÐ·ÑĥÐ»ÑĮÑĤÐ°ÑĤ: ÐľÐµÐ»Ð°Ð½Ð¸ Ñģ 36 Ð²Ð°Ðº ÑĥÑĤÐ¾Ñĩ-Ð¿ÑĢÐ¸Ð½Ð°Ð´Ð»ÐµÐ¶Ð½Ð¾ÑģÑĤÐ¸.

<answer>36</answer>